

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10
GRAND TOTAL:		50 Marks	

Annexure A – Class Test

- 1.Explain the concept of image enhancement in the spatial domain. Discuss how basic gray-level transformations, histogram processing techniques, and spatial filtering can be used to improve the quality of the captured images. Justify the suitability of each technique for this scenario.
- 2.Differentiate between image smoothing and image sharpening. Explain the working principles of spatial smoothing filters and sharpening filters. Discuss how these techniques improve image quality and explain the situations in which each technique should be applied.
- 3.Explain the image enhancement process in the frequency domain. Describe the roles of low-pass filters and high-pass filters in image enhancement. Compare their effects on image quality and discuss suitable applications for each filtering technique.
- 4.Explain the concept of image segmentation. Compare point detection, line detection, edge detection, thresholding, and region-based segmentation with respect to their working principles, advantages, limitations, and practical applications. Identify the most appropriate technique(s) for this scenario and justify your answer.
- 5.Explain the importance of edge detection, edge linking, and boundary detection in image processing. Discuss the working principles of these techniques and describe how they contribute to accurate object recognition and defect detection. Illustrate your answer with suitable examples and diagrams wherever necessary.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks	
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions		
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation		
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis		
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers		
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps		

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____